

Advait Deshmukh

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Boulder, CO, USA

RESEARCH INTERESTS

Natural Language Processing, Narrative Understanding, Computational Discourse, Interpretability

EDUCATION

Doctor of Philosophy, Computer Science

University of Colorado Boulder

Advised by Dr. Maria Antoniak

Jan 2026 – Dec 2030

Boulder, CO

Master of Science, Computer Science

University of Colorado Boulder

Advised by Dr. Maria Leonor Pacheco

Aug 2023 – Dec 2025

Boulder, CO

Bachelor of Engineering, Computer Science

Birla Institute of Technology and Science, Pilani

Aug 2017 – Aug 2021

Pilani, India

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

- [C.1] Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026. The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation. In Proceedings of the Conference on Language Modeling (COLM), Oct 2026, San Francisco, CA. Preprint coming soon.
- [C.2] Rohan Das, Advait Deshmukh, Alexandria Leto, Zohar Naaman, I-Ta Lee, and Maria Leonor Pacheco. 2026. A Structured Clustering Approach for Inducing Media Narratives. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego, CA, <https://aclanthology.org/2026.acl-long.1970/>.
- [C.3] Advait Deshmukh, Ashwin Umadi, Dananjay Srinivas, and Maria Leonor Pacheco. 2025. All Entities are Not Created Equal: Examining the Long Tail for Ultra-Fine Entity Typing. Proceedings of the 14th Joint Conference on Lexical and Computational Semantics (*SEM 2025), co-located with EMNLP 2025, <https://aclanthology.org/2025.starsem-1.15>.
- [S.1] Mreedul Gupta, Advait Deshmukh, Ashwin Umadi, Matt Pauk, and Maria Leonor Pacheco. 2026. Narrative-UFET: Narrative Generation for Ultra-Fine Entity Typing. arXiv preprint, [arXiv:2606.27598](https://arxiv.org/abs/2606.27598).

SELECTED RESEARCH PROJECTS

- **A Structured Clustering Approach for Inducing Media Narratives** Spring 2025 – Summer 2025
Collaborator BLAST Lab, CU Boulder
 - Implemented a modified DAPrompt (Xiang et al.) based relationship classifier to extract causally related event pairs across 4 datasets.
 - Infused knowledge from teacher models (Llama3, Qwen3) to aid in domain adaptation.
- **Examining the Long Tail for Ultra-Fine Entity Typing** Fall 2024 – Spring 2025
Independent study advised by Dr. Maria Pacheco BLAST Lab, CU Boulder
 - Conducted rigorous empirical analysis to study limitations of predominantly popular language model-based approaches for ultra-fine entity typing.
 - Developed proxies to predict entity frequencies based on pretrained language models and analyzed its effect on typing performance.
- **Improving LENS with Knowledge Distillation** Spring 2024
Independent study advised by Dr. Tom Yeh BAIC Lab, CU Boulder
 - Devised a selective caption filtering module on top of the approach provided by LENS (Berrios et al.) which enabled text only Large Language Models (LLMs) to "see".
 - Leveraged LLMs and knowledge distillation techniques to train a compact RoBERTa-based classifier model to streamline the filtering process.

TEACHING EXPERIENCE

- **University of Colorado Boulder** Fall 2024 – Spring 2025
Teaching Assistant, CSCI 1300: Intro to Computing Boulder, CO
 - Assisted in the course by grading, holding recitations & office hours to support students.
 - Collaborated with faculty to redesign course materials, improving student engagement.
- **University of Colorado Boulder** Fall 2023 – Spring 2024
Course Manager, CSCI 1300: Intro to Computing Boulder, CO
 - Managed operations for the course, serving over 900 students through office hours and online platform Ed.
 - Collaborated with faculty to create and grade assignments, projects & exams.

PROFESSIONAL EXPERIENCE

- **Amazon** Summer 2026
Applied Science Intern Bellevue, WA
 - Interned with Supply Chain Optimization Technologies (SCOT) on the long-term planning & forecasting team.
 - Improved evaluation for their agentic systems.
- **Societe Generale Global Solution Centre** Jul 2021 – Jul 2023
Software Development Engineer Bangalore, India
 - Developed a Retrieval Augmented Generation (RAG) based chat system to assist support teams in efficiently handling policy queries.
 - Built an innovation portal, generating 120 ideas across 7 departments to foster cross-departmental collaboration.
- **Centre for Development of Advanced Computing (CDAC)** Jul 2020 – Dec 2020
Research Intern Pune, India
 - Engineered a scalable platform for High Performance Computing (HPC) job submission to optimize data processing.
 - Developed a portal for seamless management of big data tasks, enhancing research efficiency.
- **VoiceQube** May 2019 – Jul 2019
Summer Intern Bangalore, India
 - Participated in summer internship program as part of the practice school program at BITS Pilani.
 - Developed a conversational service to assist students in their GRE preparation.

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, C++, Java
- **Frameworks & Tools:** React, Spring, FastAPI, Flask
- **Machine Learning Libraries:** PyTorch, Transformers, NLTK, Pandas, Numpy, Langchain, Ollama
- **Cloud Technologies:** Google Cloud, AWS, Microsoft Azure
- **DevOps & Version Control:** Docker, Git, Kubernetes

AWARDS AND GRANTS

- **Conference Support Funds – \$1500** Dec 2025
Computer Science Department, University of Colorado Boulder, USA
- **Travel Subsidy Recipient – \$750** Nov 2025
**SEM 2025, co-located with EMNLP 2025* Suzhou, China
- **Lloyd Botway Fellow – \$500** April 2025
Computer Science Department, University of Colorado Boulder, USA
- **Outstanding Teaching Assistant – \$500** March 2025
CSCI 1300, University of Colorado Boulder, USA
- **Winner - Poster Competition – \$500** Feb 2025
Annual Research Expo, University of Colorado Boulder, USA
- **Grad App Aid Recipient – \$500** Nov 2024
QueerInAI Boulder, USA

LEADERSHIP EXPERIENCE

- **Founder and President** Sep 2019 – May 2021
VoiceQube Pilani
 - Founded and led VoiceQube BITS Pilani Chapter to develop Voice interface-based projects.
 - Deployed BITS Buddy to handle queries about campus life, responding to ~4000 requests between January and March 2020.